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Palaeomagnetism of the dolerites of the Särv Nappe, southern Swedish Caledonides

GÖRAN BYLUND and OLLE ZELLMAN

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The dolerites of the main part of the Särv Nappe contain, after progressive alternating field and thermal demagnetization, a stable magnetic component of decl. 21.4°, incl. −9.1°, $\alpha_{95} = 12.4°$, $k = 11.3$. This gives a geomagnetic pole at lat. 21°N, long. 170°E which is in agreement with published Silurian–Devonian poles from Fennoscandia. This together with a simple 'fold test' implies that the Proterozoic dolerites became remagnetized during the Caledonian orogeny. The magnetic directions from the dolerites in the northern part of the nappe, the Ottfjället block, indicate that this area has rotated ca. 100–110° counterclockwise in relation to the rest of the nappe. □ *Palaeomagnetism, pole positions, remagnetization, rotation, dykes, tholeiitic dolerite, Särv Nappe, Tossåsfjället Group, Caledonides, Caledonian Orogeny, Proterozoic, Silurian, Devonian, western Jämtland, west central Sweden, N6229 N6321 E1392 E1220.*

Göran Bylund and Olle Zellman, Avdelningen för mineralogi och petrologi, Geologiska institutionen, Sölvegatan 13, S-223 62 Lund, Sweden; 5 January 1980.

Palaeomagnetism has had its main success in establishing the presence of large-scale movements in the Earth's crust but the method is also applicable in many other fields such as the study of events in a period of orogeny. For example, the measurements of the natural remanent magnetisation (NRM), can give information about the magnetization build-up in relation to the metamorphic history of different units, including studies of relative movements. The Geotraverse, from Östersund in Sweden across the Scandinavian Caledonides to Trondheim in Norway (Annersten 1973; Bylund et al. 1976) presents numerous possibilities for studies of this kind, and an obvious subject for such work is the well preserved parts of the Särv Nappe with its conspicuous swarm of tholeiitic dolerite dykes. This paper reports on the magnetic properties of these rocks.

Geology

The Särv Nappe is the lowermost unit in the Western Nappe Complex; a sequence of nappes that during the Caledonian orogeny were translated over the allochtonous-autochtonous Eastern Complex and the Precambrian basement of the Baltic Shield (Gee 1975*a*, 1975*b*, 1978). The translation distance has been estimated to be at least 300 km (Gee 1978). The nappe rests with a marked discontinuity on various nappes of the Eastern Complex and on the Precambrian basement, and it is overlain by the Seve–Köli Nappe Complex (Fig. 1). The border zones are composed of distinct mylonites. The nappe itself consists of sediments, the Tossåsfjället Group (Kumpulainen 1980 [this volume]), intruded by an extensive swarm of doleritic dykes. Its estimated stratigraphic thickness is 4.5–6 km (Kumpulainen, in press). The sediments consist of feldspathic sandstones with minor inlayers of siltstone, dolomite and at least one diamictite probably of glacial origin (Röshoff 1975; Kumpulainen, in press). The sediments are deposited in alluvial and shallow marine environments (Kumpulainen 1978). The dolerites were first described by Holmquist (1894), who studied them in the Ottfjället area and accordingly named them Ottfjället dolerites. In this paper, however, the term Ottfjället dolerites is reserved for the dolerites from the type locality only, in order to avoid ambiguity; the dolerites in the Ottfjället area display NRM directions that differ from those found in dolderites of the same 'Ottfjället' type in most other parts of the nappe.

The chemistry of the dolerites implies that they are abyssal tholeiites (Solyom, Gorbatschev & Johansson 1979). However, they were not formed in an oceanic environment, their intrusion was probably connected with the early stages of the opening of the Late Proterozoic Iapetus ocean (Gee 1975*a*).

The predominant type of dolerite in the Särv

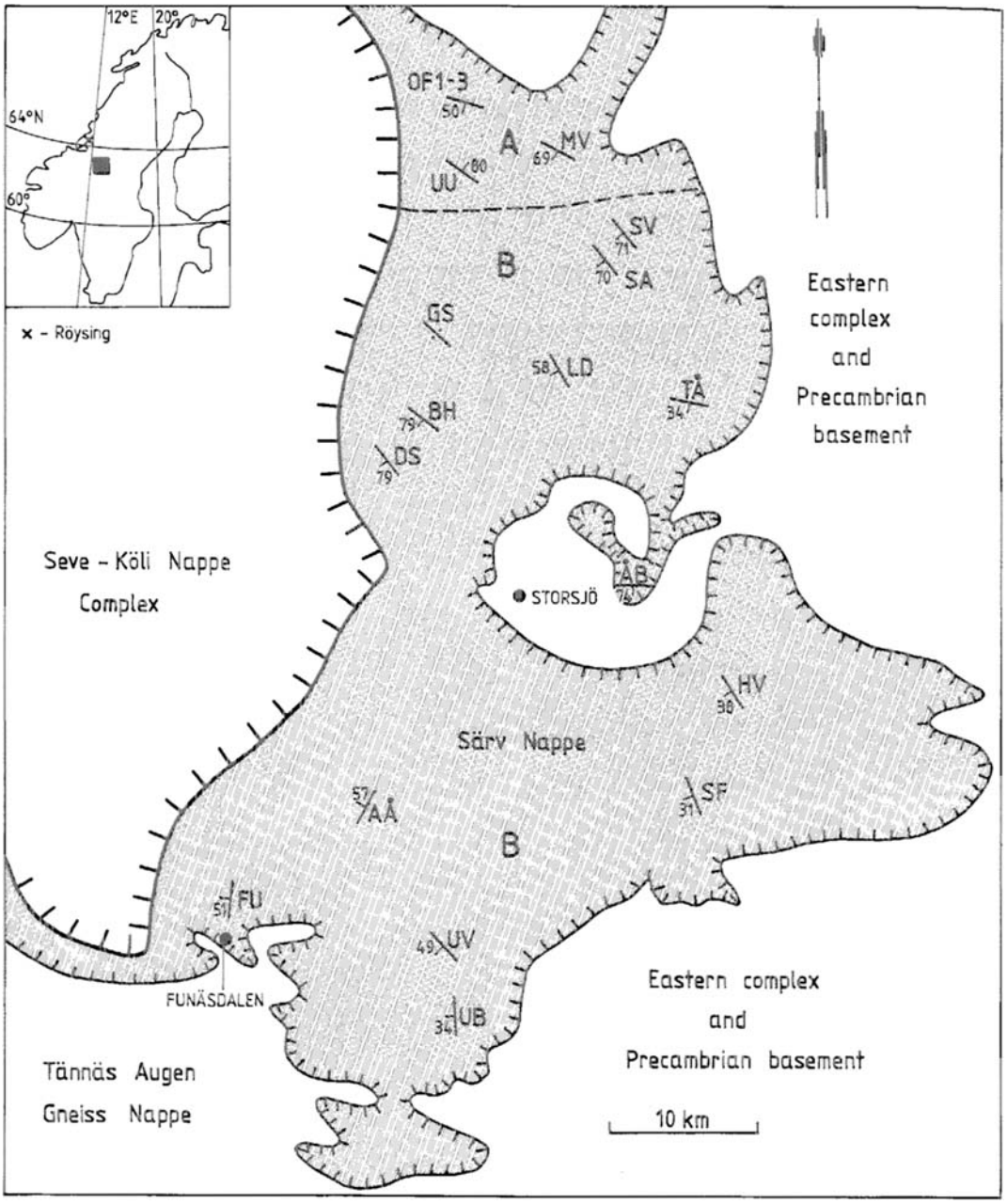


Fig. 1. Sketch map of the Särv Nappe with sampling sites and orientation of sampled dykes. Figures denote dip of the dykes. The dashed line is the approximate border between areas with different magnetic directions.

Nappe is a fine- to medium-grained porphyritic rock with phenocrysts of plagioclase in a fine-grained matrix. The contacts are clear and the dykes exhibit chilled margins. Composite dykes are present.

The Tossåsfjället Group sediments are considered to be of Late Proterozoic ('Late Precambrian') age (Strömberg 1961, p. 380), this interpretation being based on lithological similarities with the Late Proterozoic Sparagmite Group which occur locally along the Scandinavian Caledonides. The discovery of diamictites, probably of glacial origin, also favoured a Late Proterozoic age, the diamictites being correlated with the Varangian glaciation (Kumpulainen 1980 [this volume]). A Rb–Sr age of

720±255 Ma ($\lambda = 1.42 \times 10^{-11} a^{-1}$) has been presented for the dolerites of the Särv Nappe (S. Claesson 1976) and this is interpreted as the probable intrusion age. A range of K–Ar ages has been reported, from 599 to 2576 Ma (S. Claesson 1976; Point et al. 1976). It has been proposed that the dolerites contain excess argon and accordingly most of the K–Ar ages are too old (S. Claesson 1976, 1977).

The Särv Nappe rocks are metamorphosed in greenschist to lower amphibolite facies. The source for this metamorphism was considered by Gee (1975*b*) to be heat emanating from the more metamorphosed and warmer Seve Nappes that were stacked upon the Särv Nappe. The Seve Nappes were probably metamorphosed before stacking and translation begun but were a possible heat-source even after this (Gee 1975*b*). The translation of the Särv Nappe on the Tännäs Augen Gneiss Nappe was probably before the Ludlovian (Gee & Wilson 1974), it might have occurred in the Ordovician (S. Claesson 1980 [this volume]). Emplacement of the Särv and Tännäs Augen Gneiss Nappes into the province of Jämtland (Sweden) occurred in the Late Silurian to Early Devonian (Gee 1978).

Sediments similar to the Tossåsfjället Group intruded by dolerites were reported from outside the main area of development of the Särv Nappe (Gee 1977). One of these ocurrences, at Röysing (Tömmerås), 180 km NW of the main nappe, has been sampled and the results obtained are included in this study.

For a comprehensive review of the geology and petrology of the Särv Nappe lithology see Holmquist (1894), Strömberg (1961, 1969), Solyom, Andréasson & Johansson (1979) and Kumpulainen (1980 [this volume]).

Sampling and measurement

Both hand samples and cores were obtained, the latter with a portable motorized drill. Samples were orientated while still attached to the outcrop, either with a sun-compass, when weather permitted, or else with a magnetic compass. To get a test on the influence of the outcrop on the magnetic readings all sun readings were complemented with a magnetic determination. To further test for the deflection of the needle of the magnetic compass a 'check' was made by sighting on the orientation marks with the compass at least 1 m above the outcrop. Generally the differences between the different methods were less than 2° and no systematic error was observed. The sampling sites are shown on the map (Fig. 1).

For the measurements, one to four cylindrical specimens, 2.4 cm in height and 2.5 cm in diameter were obtained from each sample. The NRM was measured on a 85 Hz UGF-JR 3 or a 7 Hz Digico spinner magnetometer. Alternating field (AF) and thermal demagnetization techniques were used to remove unstable soft magnetization and to isolate the characteristic stable magnetic component. In both methods the demagnetization was done in fields of less than 10 nT. Pilot specimens from each site were step-wise demagnetized to peak fields of 100 mT. Generally stable end-points for the magnetic vectors were obtained in the range 30–40 mT, an exception being site SF for which a field of 80 mT was necessary to reach stability. After the pilot test the remaining specimens were demagnetized in five steps to 50 mT, and for site SF in eight steps to 100 mT. Thermal demagnetization was done in 13–14 steps to 600–650°C on specimens from each site. All five samples from a site close to the western peak on mount Ottfjället proved unstable during progressive demagnetization and the results from this site were rejected.

An estimate of the Curie temperature of the magnetic minerals was achieved by measuring the saturation remanence (J_s) through a range of temperatures. This was done by weighing chips and powder samples, approximately 0.4–1.0 g, while they were hanging in a strong magnetic gradient. A saturation field of 200–300 mT was used and the measurements were made in air.

The results are listed in Table 1, and examples of the behaviour of pilot specimens during demagnetization are presented in Fig. 2. Sample averages are used for calculation of site means and Fisher's (1953) statistics used for calculating mean directions.

The behaviour of the dolerite specimens during the stepwise AF-demagnetization varied and different patterns are presented in Fig. 2A. Small changes during the treatment are exemplified by the tight cluster of points for sample OF 62. More common was the situation where a soft component is removed in a field less than 20 mT (SV 15). Sample HV 8 represents a case where a component aligned along the direction of the Earth's present field is removed. The AF-demagnetization curves from the dolerite samples show that a peak field of about 30 mT removes 50% of the NRM and that ca. 10% is left after 100 mT. For comparison, sediment specimens were given the same treatment, an example being presented in Fig. 2A where ÅB 37 is a feldspath-

Table 1. Magnetic directions after AF-demagnetization. N = number of samples; AF = demagnetizing field with least scatter in directions and presented in this table; α_{95} = semi – angle of the 95% cone of confidence; k = the best estimate of Fisher's precision parameter.

Site	N	AF (mT)	Decl. (°)	Incl. (°)	α (°)	k
Group A						
Ottfjället, OF 1	14	20	274.4	17.8	11.8	12.2
Ottfjället, OF 2	7	30	279.3	22.0	28.9	5.3
Ottfjället, OF 3	10	30	267.4	13.9	21.7	5.9
Ottfjället, mean, sites	3	–	273.9	18.0	–	–
Middagsvalen, MV	13	40	297.6	9.1	15.1	8.4
Uluberget, UU	9	30	307.6	35.8	26.2	4.8
Group B						
Skraupvalen, SV	12	40	31.9	–2.7	14.6	9.8
Santa, SA	9	30	24.9	–3.3	20.7	20.7
Åviksberget, ÅB	9	40	24.7	–9.4	9.5	30.3
Bockhammaren, BH	11	40	3.6	25.2	27.8	3.7
Gråsjön, GS	11	40	355.3	–18.8	19.5	6.4
Dunsjöfjället, DS	10	40	9.9	–8.7	12.3	16.4
Tossåsfjället, TÅ	19	30	31.9	–6.2	12.9	7.7
Lunndörrspasset, LD	13	30	21.5	11.4	12.7	11.6
Henvålen, HV	5	40	35.7	–1.5	13.7	32.2
Uggvallen, UV	9	40	345.8	–31.6	39.3	2.7
Anåfjället, AÅ	5	40	37.3	–8.4	9.2	69.6
Ulvberget, UB	7	50	24.5	–11.8	8.1	56.5
Särvfjället, SF	4	80	20.2	–2.6	9.6	92.6
Funäsdalsberget, FU	17	40	62.8	–50.1	21.1	3.8
Mean, sites	14	–	21.4	–9.1	12.4	11.3
Röysing, RÖ	7	40	16.0	–20.3	12.1	25.5

ic sandstone collected 0.5 m from a dolerite contact. It lost only 10% in its intensity up to 10mT but showed no further decay up to 100 mT. This indicates a stable magnetic component with high coercivity in contrast to the dolerites with their medium to low coercivity.

The thermal demagnetization presented the same directional patterns as the AF-treatment although remanence directions for the dolerites became erratic at 600°C. The demagnetization curves, Fig. 2D, are of the 'shoulder' type indicating a single blocking temperature interval of 550–600°C for the dolerites and a slightly higher for the sandstone. This suggests that magnetite or a titanomagnetite close to pure magnetite in composition is a carrier of the magnetic remanence of the dolerites. For the sandstone the higher blocking temperature and the hard coercivity implies that hematite is a major remanence carrier. This conclusion is further substantiated by the J_s/T curves of Fig. 3. In the dolerite a Curie temperature close to that for magnetite, 578°C, is found while the sandstone sample (ÅB 37:3) displays a more complex J_s/T pattern indicating Curie temperatures close to 400, 600 and 700°C respectively. The latter two inflexion points indicate magnetite and hematite while the first may be attributed to maghemite or pyrrhotite.

The mean initial NRM was 1.8 Am^{-1}, range 15–0.01 Am^{-1}, measured on 239 specimens. Mean volume susceptibility was 13.4×10^{-3} (SI-units), range $32–0.5\times10^{-3}$, measured on 43 specimens. The Q-value (Koenigsberger ratio) was generally above unity, the mean value was 2.8 and the range 10.09–0.04.

Ore microscopy

In general, the stable NRM in most rocks is carried by grains of single- or pseudo-single domain size, which for magnetite generally means grains below the resolving power of a standard optical microscope and consequently any conclusions drawn from a microscopic study will be of general nature. A Q-value above unity, as is the case for the dolerites of the Särv Nappe, generally indicates that the stable NRM is carried by very small grains (Stacey & Banerjee 1974, p. 115).

The main opaque mineral in the dolerites is ilmenite, usually in rounded or skeletal grains or together with magnetite as exsolution lamellae

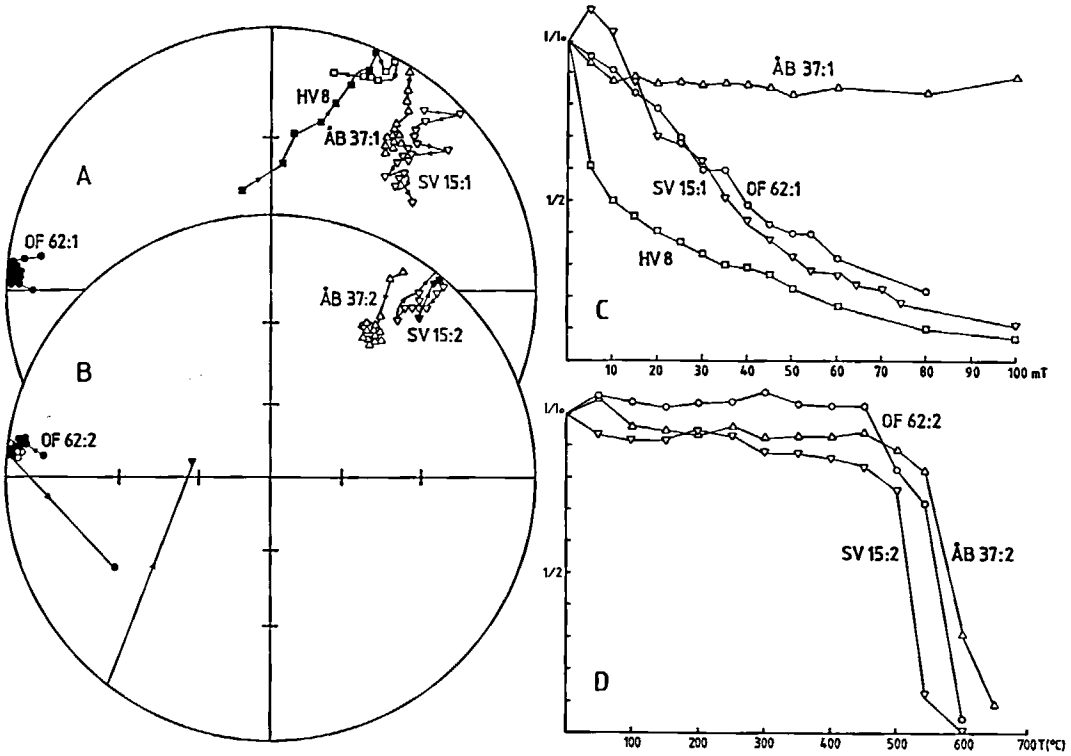


Fig. 2. Behaviour of pilot specimens during AF (A, C) and thermal demagetization (B, D). Sample ÅB 37 is a feldspathic sandstone, the others are all dolerites. Closed symbols in A and B denote inclination pointing down (positive) and open symbols pointing up (negative). C and D show normalized curves for the magnetic intensity during demagnetization.

from titanomagnetite. The ilmenite is often altered to leucoxene. Single grains of magnetite occur. Sulphides are common, mainly as idiomorphic pyrite but pyrrhotite and chalcopyrite are also found. Chromite has been observed. Fine grains or dust of opaques, probably magnetite, occur as alteration products in pseudomorphs of olivine and augite.

Minute grains of hematite together with some magnetite and sulphides are found in the sandstone samples studied. This confirms the suggestion that at least part of the NRM in the sediments is carried by magnetite and hematite.

Discussion of directions of magnetization

The within-site scatter in directions is large for some sites, e.g. OF 2, UU. BH, and UV as evidenced by α_{95} larger than 25° (Table 1). However, their mean directions have a similar trend to that which is outlined below and thus they are included in the discussion.

Two groups of stable magnetic directions emerge after progressive demagnetization of the dolerites (Fig. 4). One, called group A, has a shallow downward direction pointing WNW, the other, group B, a shallow upward direction towards NNE.

The A directions are all from sites situated in the northern part of the Särv Nappe; three sites on Mount Ottfjället (OF 1–3) and sites on the mountains Middagsvalen (MV) and Uluberget (UU). Remaining site directions form the B-group. The difference between the mean remanence of the Ottfjället sites in the A-group and the B-group mean is 111°. Between sites MV and UU in the A-group and the B-group mean, the differences are less. This supports Strömberg's (1961) suggestion that the Ottfjället area has rotated counter-clockwise in relation to the main, southern part of the nappe. The results also imply that the Middagsvalen and Uluberget

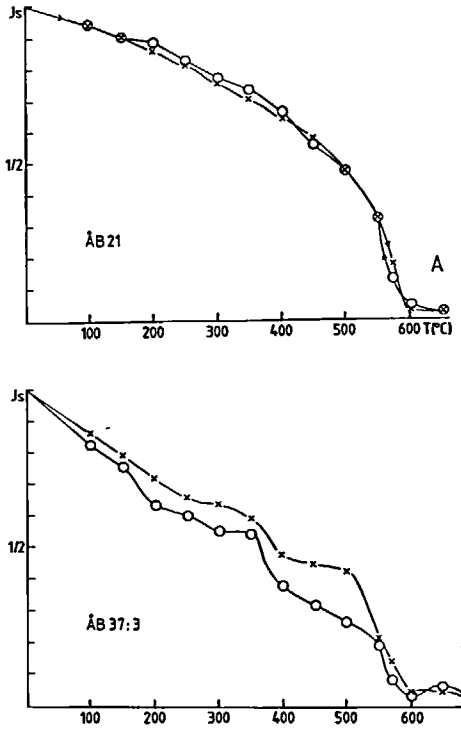


Fig. 3. Saturation magnetization (J_s) *versus* temperature (T) for dolerites (A) and feldspathic sandstone (B). × = heating; O = cooling. The scale on the J_s-axis is arbitrary.

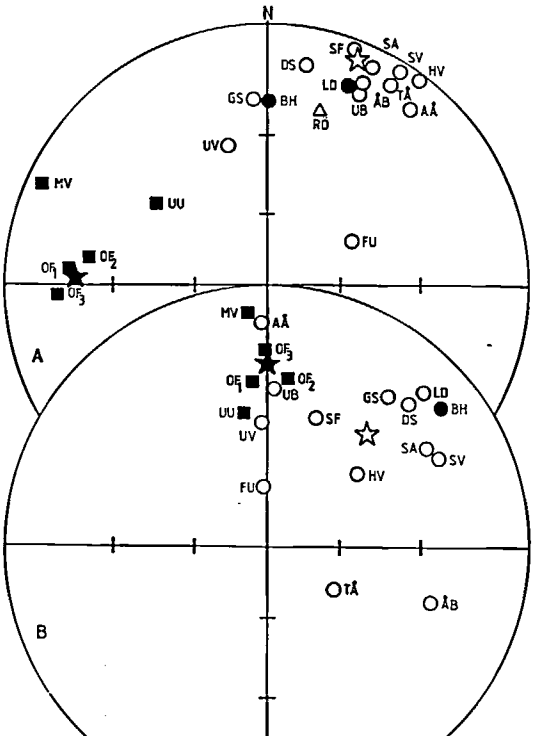


Fig. 4. Site mean directions after demagnetization, before (A) and after (B) reorientation. Wulff net projection. Symbol conventions as in Fig. 2. ■ = sites from the Ottfjället block, group A; O = sites from the main part of the nappe, group B; Δ = the Röysing site. Closed star denotes the mean for the Mount Ottfjället sites in A and all group A sites in B. The open star denotes the group-B mean.

areas have been involved in this rotation but to a slightly lesser extent. Strömberg based his interpretation mainly on the difference in dyke orientation between the E–W-trending dolerites on Mount Ottfjället and the generally NNW–SSE-trending dykes in most other parts of the nappe and he mentioned a figure of 70° for this rotation. He also suggested that an area on the eastern margin of the nappe with E–W-trending dykes had been rotated in the same manner. However, two sites sampled there, Mounts Åviksberget (ÅB) and Tossåsfjället (TÅ), have stable magnetic directions that agree with the B-group directions (Fig. 4A). A simple test was made in an attempt to study this problem of dyke orientation versus remanence directions. Strömberg (1961, p. 76) implied that the dykes intruded vertically, an idea based on the high angle between the dykes and the bedding planes of the sediments. This suggests an apparently parallel and vertical mode of intrusion. This justifies a test whereby the dykes are reorientated to a common direction, in this case N–S, with vertical dip and to rotate the remanence directions accordingly (Fig. 4B). After this operation the B-group directions disperse, and the precision parameter k (Fisher 1953) decreases from 12.4 to 4.9. As the dykes generally tilt in the same direction, a fold test (Graham 1949; McElhinny 1964) is not significant. The decrease in precision is however an indication that a new magnetization has been imposed on the dolerites after their intrusion. If so, this remagnetization is older than the rotation of the Ottfjället area, group A directions, but younger than a possible rotation of the area E of Storsjö with sites ÅB and TÅ. This assumes that the dolerites acquired their magnetism at the same time.

After the realignment the B-group directions split into three diffuse parts, one comprising the sites FU, UV, AÅ, SF, and UB, all from the southern part of the nappe and with a north-

Table 2. Early–Middle Palaeozoic pole positions from Scandinavia.

Rock unit	Pole position		Age	Reference
	lat. °N	long. °E		
1. Fen alkaline complex	20	162	Early Cambrian	Storetvedt 1973
2. Sulitjelma gabbro	10	177	Ordovician	Piper 1974
3. Ringerike sandstone	21	159	Late Silurian	Storetvedt et al. 1968
4. Limestones, Gotland	20	170	Late Silurian	C. Claesson 1979
5. Kvamshesten Old Red Sandstone	22	170	Middle–Late Devonian	Lie et al. 1969
6. Röragen, Old Red Sandstone	19	160	Early Devonian	Storetvedt & Gjellestad 1966
Särv Nappe Dolerites	21	170	Proterozoic	This paper

trending direction, a second group of site directions from the central and northern part of the B area and trending northeast, and as a third group the sites ÅB and TÅ from the eastern fringe with directions pointing towards east. The A-group directions on the other hand came closer together after the reorientation. It is not possible to discern if those changes between directions within each group are due to rotations within the nappe during translation or to differences in the original orientation of the dykes. It is to be expected when considering, as in this case, a sheet of rocks of ca. 100×300 km in area and only a few kilometres in thickness which has been transported at least 300 km that it has suffered internal disturbances. This is also evidenced by zones of mylonites within the nappe and the tilting and gentle folding of dykes. Considering this fact and the rather good grouping of the B-group directions before the test it is likely that a remagnetization of the dolerites of the Särv Nappe has occurred.

Further evidence for the remagnetization theory is the good agreement between published Silurian–Early Devonian poles from Scandinavia (Table 2 for references) and the Särv Nappe B-group pole. This is situated at lat. 21°N, long. 170°E, $d_p = 6.3°$, $d_m = 12.5°$ (Table 2 and Fig. 5). The group A pole is situated at lat. 10°S, long 284°E (Mount Ottfjället sites only).

It has to be considered if the dolerites retain a Late Proterozoic magnetism. There is, however, a lack of Fennoscandian palaeomagnetic data for the relevant period, ca. 700–750 Ma, to use for comparison and this leaves a possibility that the group A pole represents the Late Proterozoic geomagnetic field. Further there is a possibility of superposition of A and B components, specially for sites with large scatter and a position intermediate between the two groups. In Fig. 6 are plotted all initial NRM directions from four of those sites, UU, UV, GS and BH, together with the end-point at the demagnetization level with least scatter for the site. No smearing of directions from one group to another is observed, neither is any other prominent direction found.

The metamorphic grade of the Särv Nappe rocks, the tectonic evidence given by the 'fold' test and the Silurian–Early Devonian pole position persuade us that a model of remagnetization of the Särv Nappe rocks accompanied by internal movements of blocks within the nappe is to be preferred.

Abrahamsen et al. (1979) report a similar phenomenon from the Fongen–Hyllingen gabbro

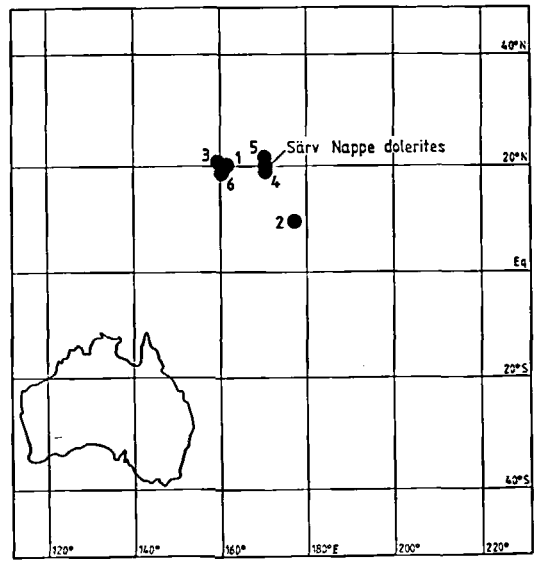


Fig. 5. Cambrian–Devonian palaeopoles from Fennoscandia. Figures refer to Table 2.

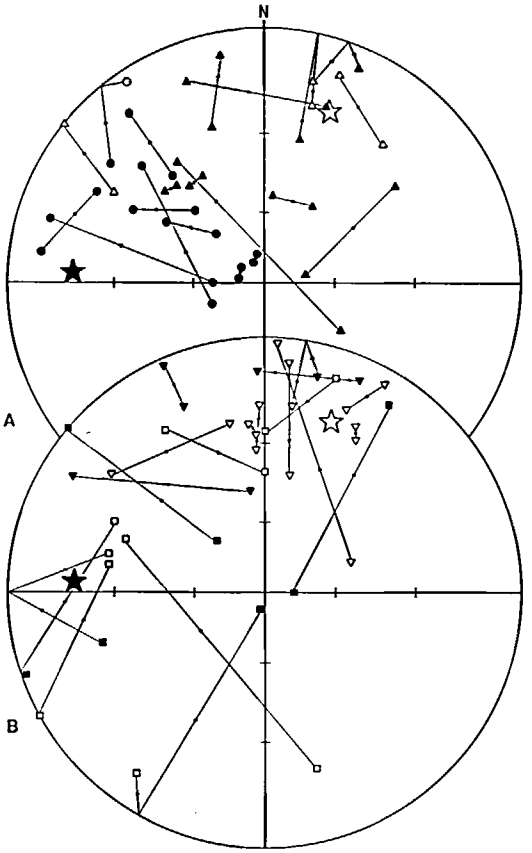


Fig. 6. Magnetic directions before and after demagnetization. In A, O = site UU, Δ = site BH. In B, □ = site UV and ▽ = site GS. Closed symbols denote positive inclination. Filled star is the Mount Ottfjället mean direction, open star the group-B mean direction.

complex situated 75 km SE of Trondheim. The gabbro contains two magnetic components of which one is of Late Silurian–Early Devonian age while the other is situated ca. 90° away. Their interpretation is that the latter pole either represents a late Caledonian geomagnetic excursion of considerable duration or a 90° rotation of a crustal block within the Scandinavian Caledonides. The geomagnetic excursion theory is not applicable as an explanation for the magnetic differences within the Särv Nappe as all rocks within the nappe would have been affected in the same way.

The event that caused the remagnetization of the Särv Nappe rocks is very likely to have been the regional metamorphism of the rocks in greenschist to lower amphibolite facies. The thermal demagnetization tests on the dolerites indicated a discrete blocking temperature interval of 550–600°C. This is the temperature range in which the magnetization will be effectively fixed, and it is valid for the time the specimens were exposed to this temperature in the laboratory test, ca. 300–1000 s. However, the effective blocking temperature will be considerably lowered during slow cooling. Considerations of the effects of slow cooling and thermal remanent magnetization (TRM) have been made by Pullaiah et al. (1975), who showed that, theoretically, the remanence in magnetite with blocking temperatures within 50°C of the Curie point can survive lower amphibolite facies metamorphism. However, in the same paper they presented experimental data that were at variance with their theory and they concluded that very little, if any, remanence in magnetite can survive high greenschist facies metamorphism. This indicates the possibility of at least partial removal of the original TRM in the dolerites and a build up of a new TRM, or a possible chemical remanent magnetization (CRM) due to the formation of new magnetic minerals. It is difficult to divide the remanence into components which could represent different magnetization events because the carriers of the stable remanence are magnetic grains with a single blocking temperature interval. The soft magnetization that is removed during the first steps of demagnetization can probably be attributed to the Earth's present field or local fields in the laboratory and carries no information on the ancient geomagnetic field. This process of acquiring a new magnetization probably covered the period from when the heat-generating Seve–Köli Nappe Complex was stacked upon the Särv Nappe rocks and the metamorphism of the latter began until the whole package was translated to its present position. Further, the burial of the Särv rocks under several kilometres of Seve–Köli rocks was a heat source in itself which prolonged the cooling of the Särv Nappe. This series of events comprises most of the Caledonian orogeny in this part of the Scandes but it is to be assumed that for the acquirement of a new magnetism the most important period ought to have been the initial stages of this sequence. According to Gee & Wilson (1974) the culmination of the orogeny in this area was in the Middle to Late Silurian but the Särv nappe rocks may have been metamorphosed in the Ordovician (S. Claesson 1980 [this volume]). The radiometric age-determinations give no clue to the date of the magnetization; the Rb–Sr age of 720 ± 225 Ma is the intrusion age and dates the obliterated original TRM, and the K–Ar ages are too scattered to help solving the problem.

The dolerite from Röysing, in the Tömmerås area, which is very similar to the dolerites of the Särv Nappe (Andréasson et al. 1979) and intrudes sediments of similar type, the Leksdalsvann Group (Springer Peacy 1964; Gee 1977), has a stable magnetic direction similar to that of the Särv B-group (Fig. 4A). This indicates that the Röysing dolerite has a magnetic history similar to that of the dolerites of the Särv Nappe in Sweden.

The stable remanence directions found in the measured sediments (11 samples) varied; a few had the same directions as the dykes from the same site, e.g. ÅB 37 (Fig. 2 A–B) whilst others showed diverging directions. It is possible that a comprehensive study of different types of sediments in the Särv Nappe might shed more light on the number (one or more) of magnetizations that are recorded and their relative age. The opportunity to find rocks with magnetic minerals which have different blocking temperatures will make it possible by vector subtraction to separate the different directions. Further chemical demagnetization can be used on the reddish sandstones commonly found in the nappe. This is a method which has proved successful in separating different magnetic directions and such work is in progress.

Conclusions

The original intrusional TRM of the Late Proterozoic dolerites of the Särv Nappe became obliterated or overprinted during the culmination of the Caledonian orogeny in this part of the Scandes. This occurred in the Ordovician or Silurian when the Seve–Köli Nappe Complex was stacked upon the Särv Nappe and translation began from a site west of the present coast of Norway. The movement was accompanied by tectonic disturbances within the Särv Nappe and the palaeomagnetic results indicate that those internal movements were contemporary with or occurred before the dolerites acquired their present magnetization. One exception is the northern part of the nappe, the Ottfjället block, which evidently rotated ca. 100–110° counterclockwise relative to the main part after the period of remagnetization.

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A Swedish contribution to the International Geodynamics Project

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